

6(a)	(both have) radial field lines	B1
6(b)(i)	2.1 cm	B1
6(b)(ii)	$E = \frac{Q}{4\pi\epsilon_0 r^2}$ e.g. $r = 2.1 \text{ cm}$, $E = 1.30 \times 10^5 \text{ V m}^{-1}$ $Q = 4\pi\epsilon_0 r^2 E$ $= 4 \times \pi \times 8.85 \times 10^{-12} \times 0.021^2 \times 1.30 \times 10^5$	C1
	$= 6.4 \times 10^{-9} \text{ C}$	A1

Question	Answer	Marks
6(c)	$C = \frac{Q}{V}$ <i>either</i> $V = \frac{Q}{4\pi\epsilon_0 r} \text{ leading to } C = 4\pi\epsilon_0 r$	C1
	$C = 4 \times \pi \times 8.85 \times 10^{-12} \times 0.021$	C1
	$(C =) 2.3 \times 10^{-12} \text{ F}$	A1
	<i>or</i> $V = \frac{Q}{4\pi\epsilon_0 r}$ $= \frac{6.4 \times 10^{-9}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.021}$ $= 2740 \text{ V}$ $C = \frac{6.4 \times 10^{-9}}{2740}$	(C1)
	$= 2.3 \times 10^{-12} \text{ F}$	(A1)